

Recall that when you visit a URL, your web browser is opening up a network connection to a web server. It knows which web server based on the host portion of the URL (known as the *domain*). The language that the browser uses to speak to the web server is called the *HTTP protocol*; it's a set of rules specifying how the browser and web server communicate. The HTTP protocol is split into two parts: the request from the browser to the server and the response from the server to the browser. Django encapsulates both of these in objects, helpfully named `django.http.HttpRequest` and `django.http.HttpResponse`.

A function-based view has two requirements: it takes at least one argument, an `HttpRequest` object, and returns an `HttpResponse` data structure (or something like it). Django uses the `HttpResponse` to produce the response document, including headers and HTML body for your browser.

By convention, the `HttpRequest` argument gets called `request` and contains many of the headers your browser sent the web server, including the URL. Your view function can take more arguments than just the request. Django supports a way of specifying multiple parameters inside a URL and mapping these to your function arguments. It also supports query parameter—that's everything after the question mark (?) in a URL. Query parameters get embedded in the request object. You'll learn more about both these techniques in future chapters.

To keep it simple, your Credits page will contain your name and Nicky's, and not much else. To keep it really simple, it will be plain text, rather than HTML. Your view is only a few of lines of code: a function that takes a `request` object, declares a string, and returns an `HttpResponse` object. Figure 2.4 shows the files that need to be changed to declare your view.

The `HttpResponse` you return needs two arguments: the content to return, and the type of that content. The content, in this case, is a string containing your credits. The second argument is `content_type`, and it changes the MIME type of the response.

Web browsers are capable of displaying all kinds of different content: HTML, images, CSS, and more. An HTTP response can contain a MIME-type header that indicates just what the content is. MIME stands for *multipurpose internet mail extensions* and, as the name implies, originally came about to define how to attach different kinds of content to an email message. By default, the `HttpResponse` object uses a MIME type that indicates HTML, as the credits page returns plain text, the `HttpResponse` object needs to be told this by setting `content_type="text/plain"`. To write your first view, open `RiffMates/home/views.py`, and replace the stub code with the content of the following listing.

Listing 2.2 Credits view

```
# RiffMates/home/views.py
from django.http import HttpResponse
```

← Import the `HttpResponse` object Django uses to respond to web requests.